

A Circumplex Model of Affect

Russell (1980), Journal of Personality and Social Psychology 39(6) · tags: valence, arousal, stress, calm, angry, circumplex

CURATED SUMMARY — NOT a verbatim quote (full text is paywalled; this paragraph restates well-established, widely-cited facts about the paper, not the paper's original sentences)

Russell proposed that affective states are not independent categories but are systematically interrelated, and can be represented as points on a circle in a two-dimensional space defined by two bipolar axes: pleasure–displeasure (valence) and arousal–sleepiness (activation).

Eight affect terms are placed evenly around the circle at roughly 45-degree intervals in the order: pleasure, excitement, arousal, distress, displeasure, depression, sleepiness, and relaxation — so, for example, 'excited' sits between high arousal and high pleasure, while 'distressed' sits between high arousal and low pleasure (displeasure).

Relevance to MedOracle: this circumplex is the theoretical basis for mapping DEAP's continuous valence/arousal/dominance ratings onto this project's 5 discrete classes (stress, calm, happy, sad, angry) via quadrant thresholds.